

Supernovae: An Explosion in Data

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- 1) Introduction to supernovae and current era of “big data”
- 2) Scaling Bayesian inference techniques for next generation surveys
- 3) Developing machine learning tools to classify supernovae

Part 1: Supernovae and “Big Data”

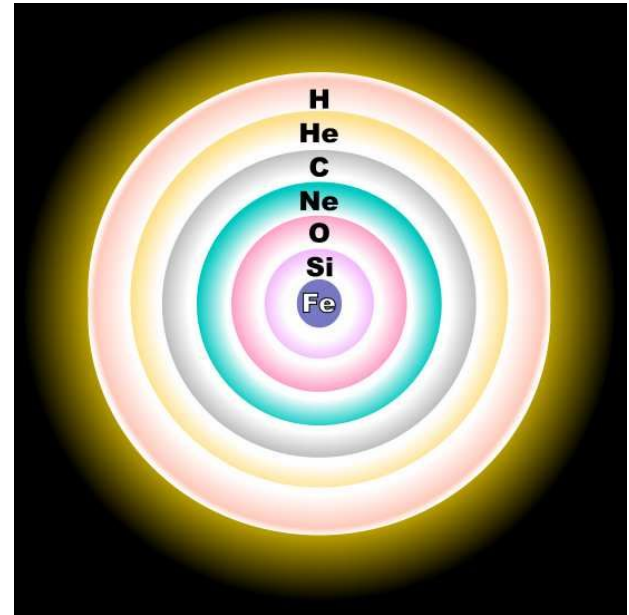
Supernovae

Type Ia



Credit: Adriana Manrique Gutierrez svcs.gsfc.nasa.gov

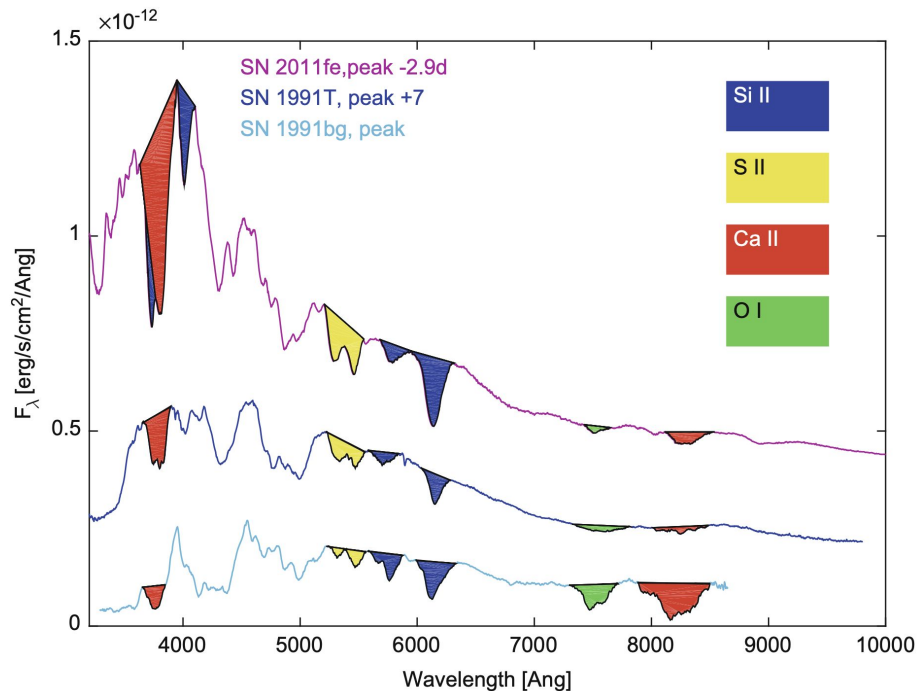
Core-collapse



Supernova Data

Spectra

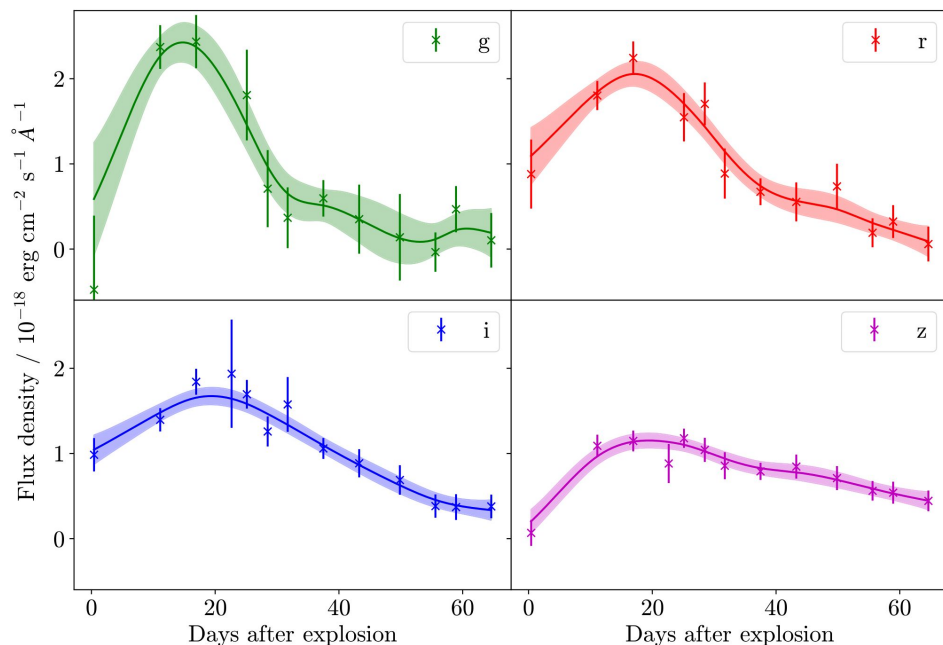
- Detailed, expensive to obtain



Gal-Yam 2017

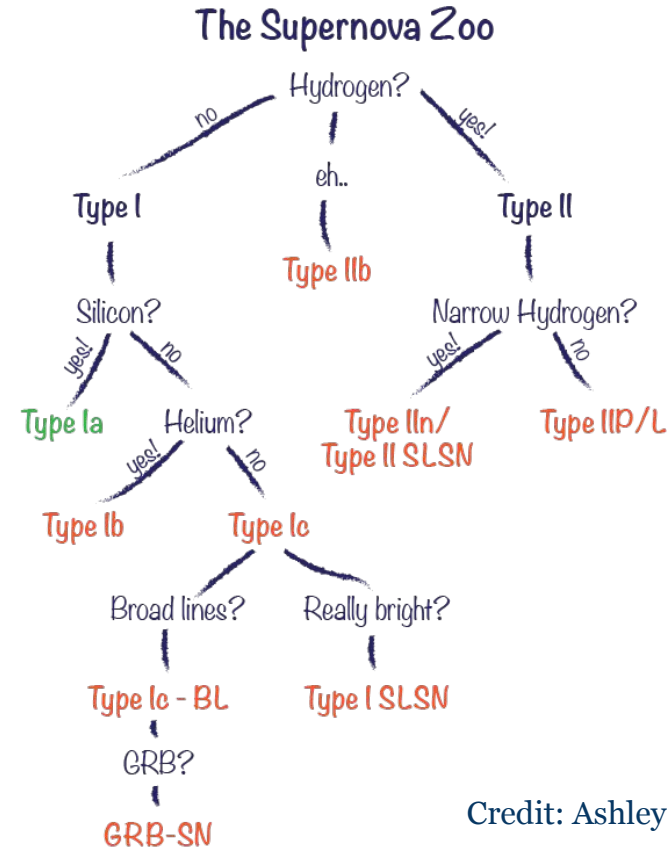
Photometry

- Easy to obtain, less detailed



Supernova Classes

- Many, many different classes
- Categorised mostly by presence of different elements



Credit: Ashley Villar

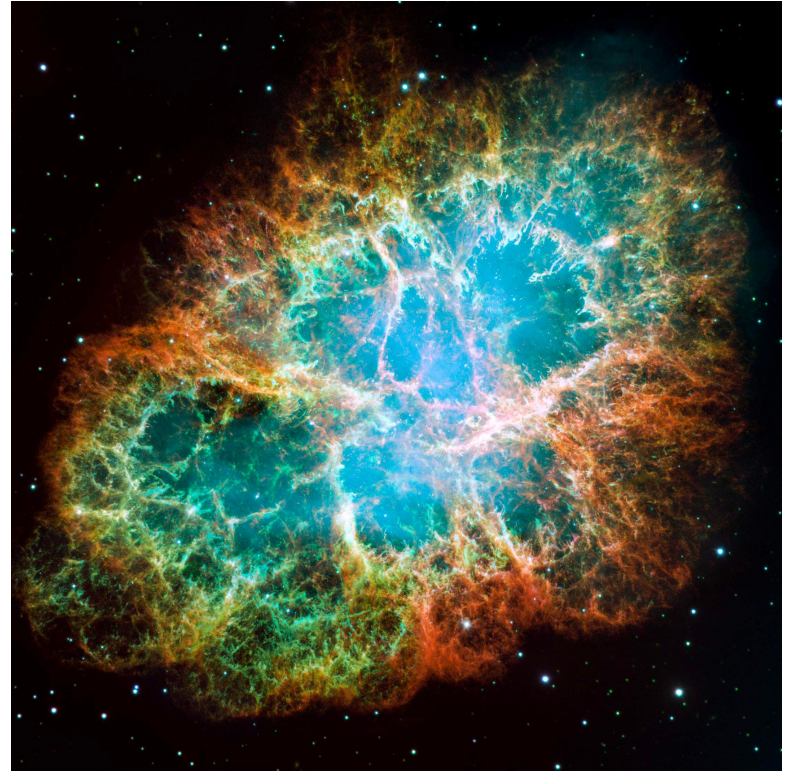
Why are supernovae interesting?

Astrophysics

- Create and distribute heavy elements
- Release energy into their surroundings
- Environment to study extreme explosion physics

Cosmology

- Type Ia supernovae can be used to measure distances



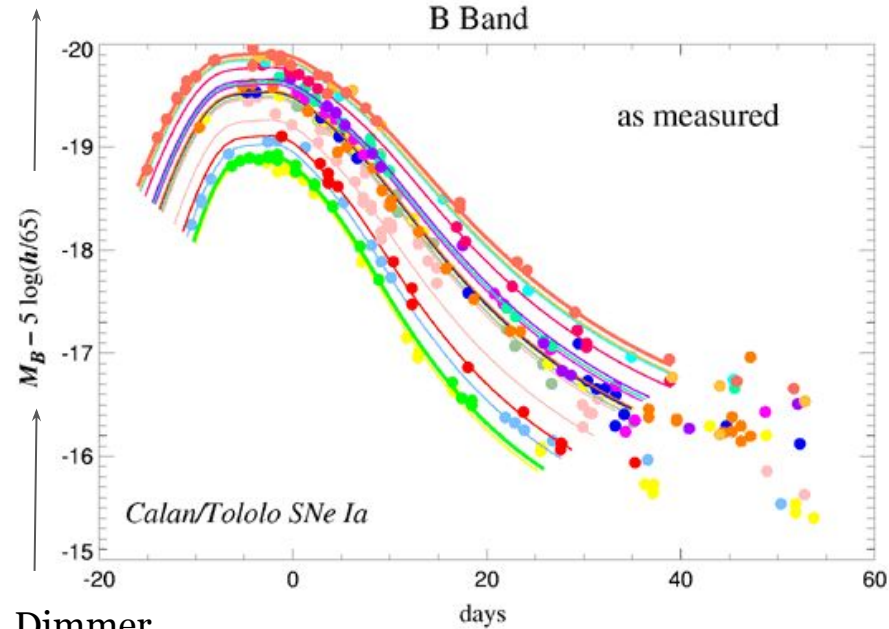
Credits: NASA, ESA, J. Hester and A. Loll (Arizona State University)

$$F = \frac{L}{4\pi d^2}$$

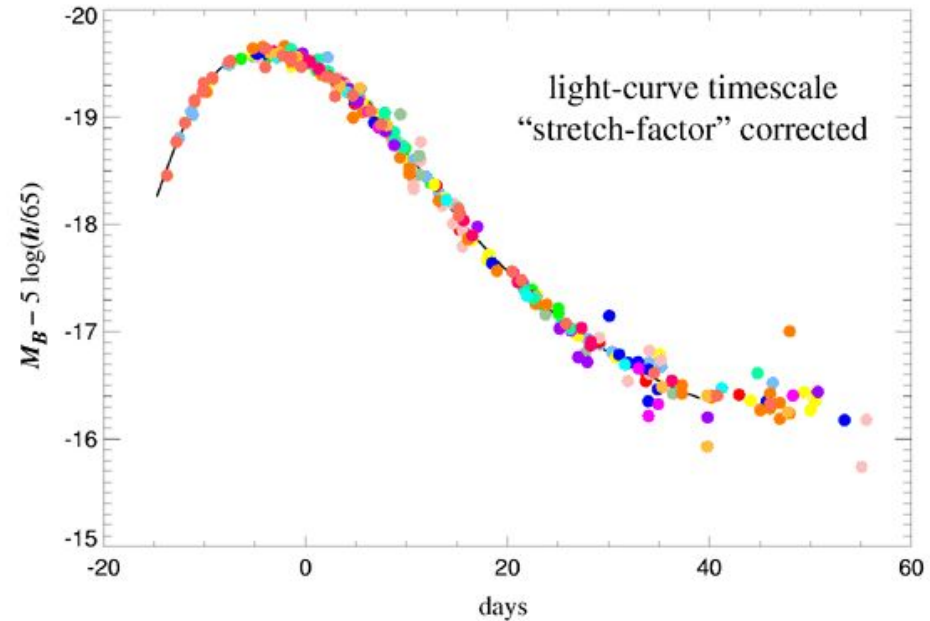
$$\mu = m - M$$

Standard(isable) Candles

Brighter

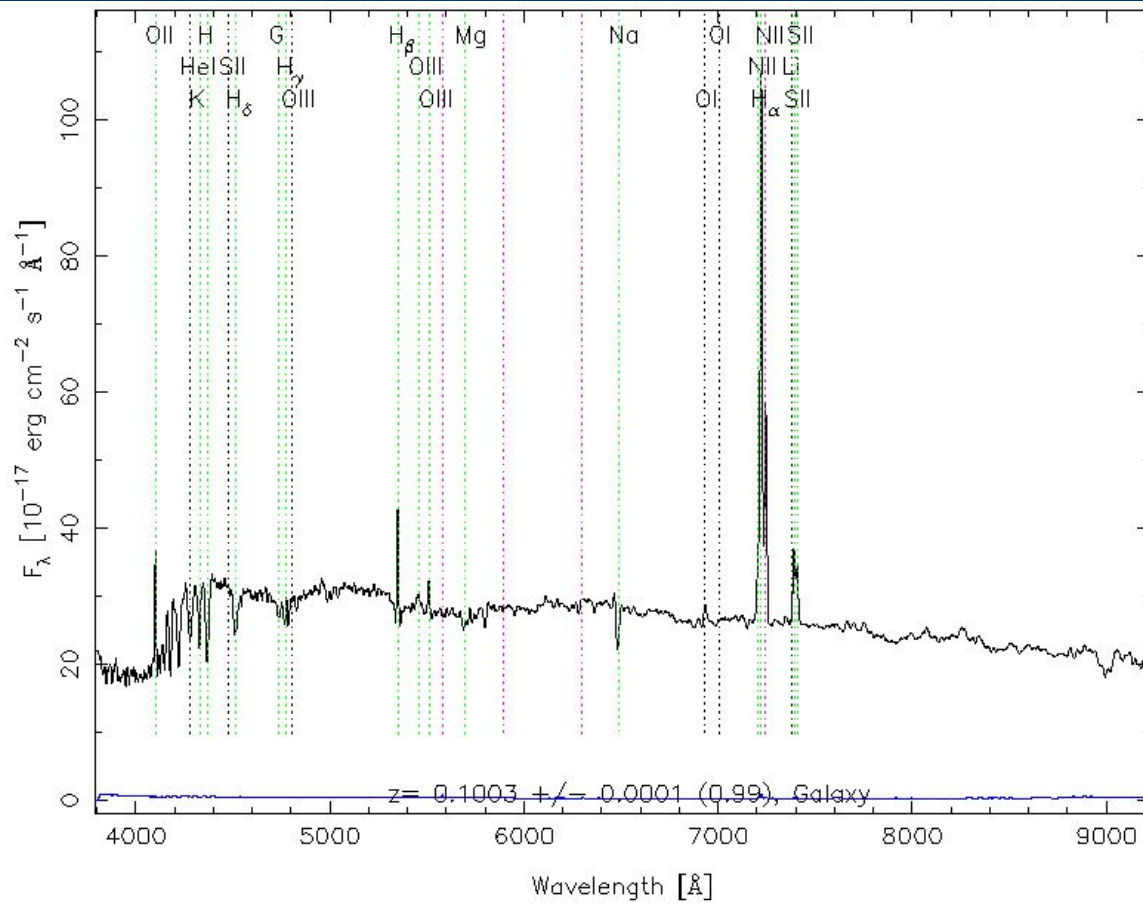


Dimmer



From Supernova Cosmology Project

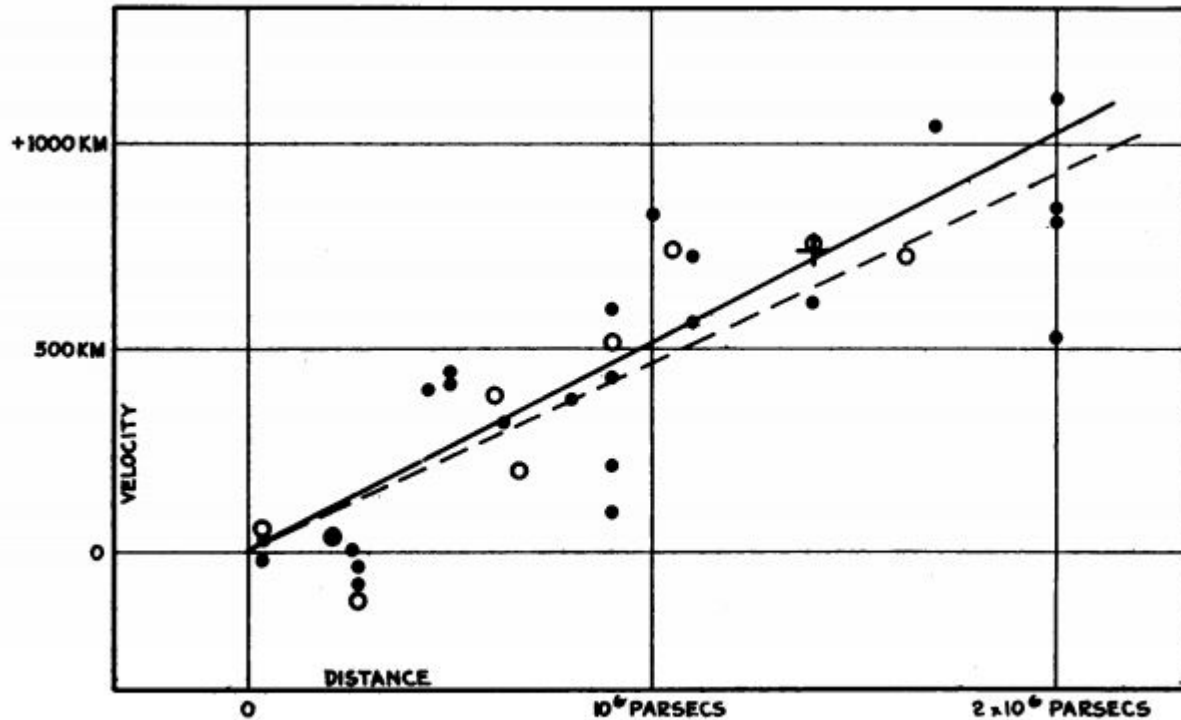
Redshift



$$\frac{\lambda_o}{\lambda_e} = 1 + z$$

***Credit: Sloan Digital
Sky Survey***

Hubble Diagram



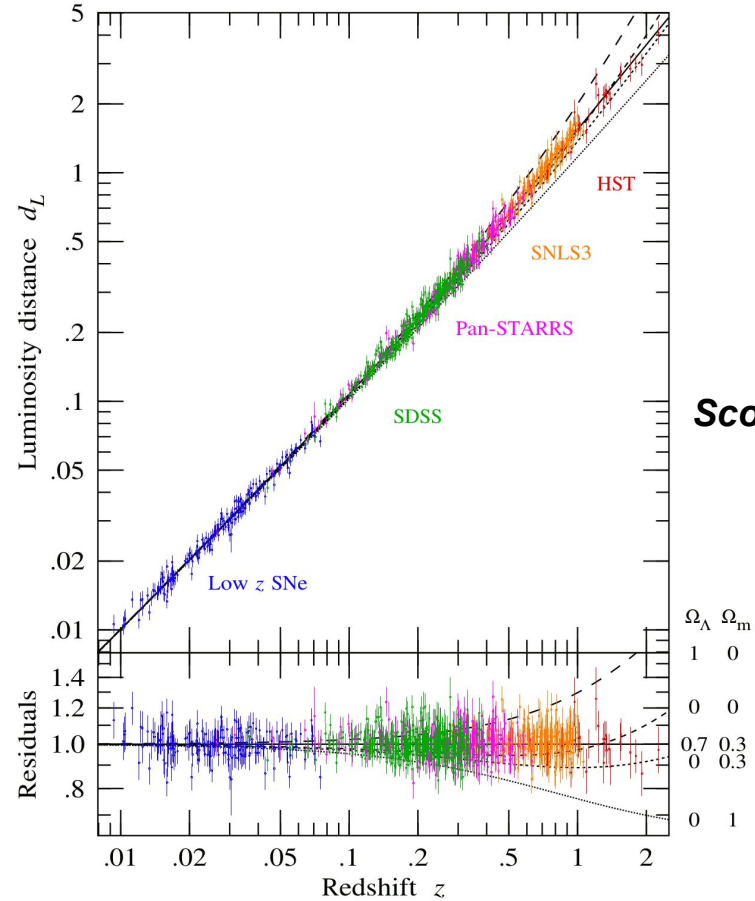
Hubble 1929

FIGURE 1

Velocity-Distance Relation among Extra-Galactic Nebulae.

Hubble Diagram

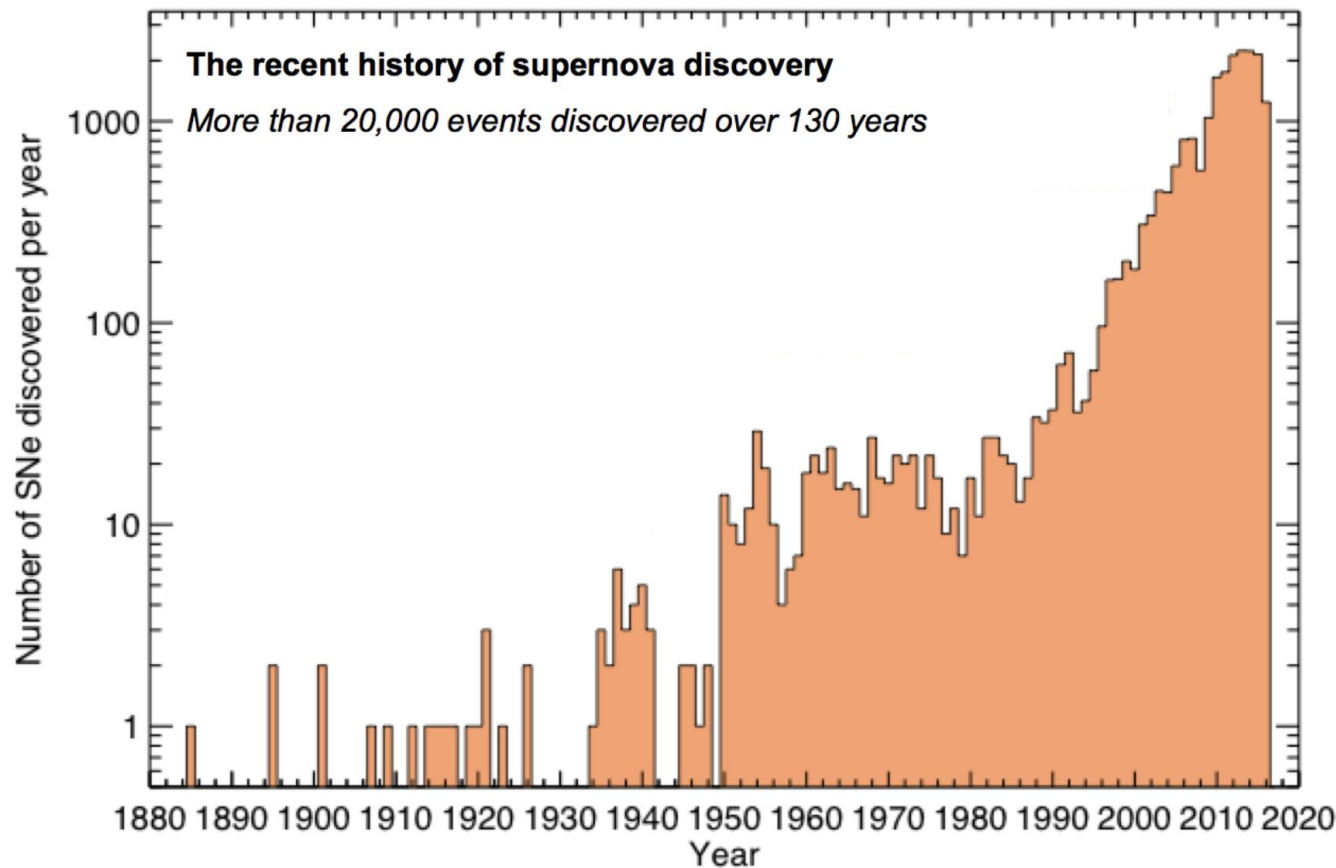
- Combining supernova distances with redshift measurements allows for inference of properties of the universe
- Analysis of type Ia supernovae first demonstrated that the expansion of the universe was accelerating



Scolnic+2018

- Type Ia supernovae
 - ‘Standardisation’ not perfect - supernovae in higher mass galaxies brighter than those in low mass galaxies
 - Not clear what progenitor systems of type Ia supernovae are - multiple routes are possible and could all occur
 - Hubble tension - discrepancy between measurement of local rate of expansion of universe between different methods
- Core-collapse supernovae
 - Many unknowns about the physical mechanisms involved
 - Unclear if different classes are part of a continuum or distinct events

The Future of Supernova Science



From 2024(ish):

Legacy Survey of
Space and Time at
Vera Rubin
Observatory (LSST) -
tens of thousands
per year

Credit: Mark Sullivan

Part 2: Scaling Bayesian Inference for Next Generation Surveys

Tripp formula

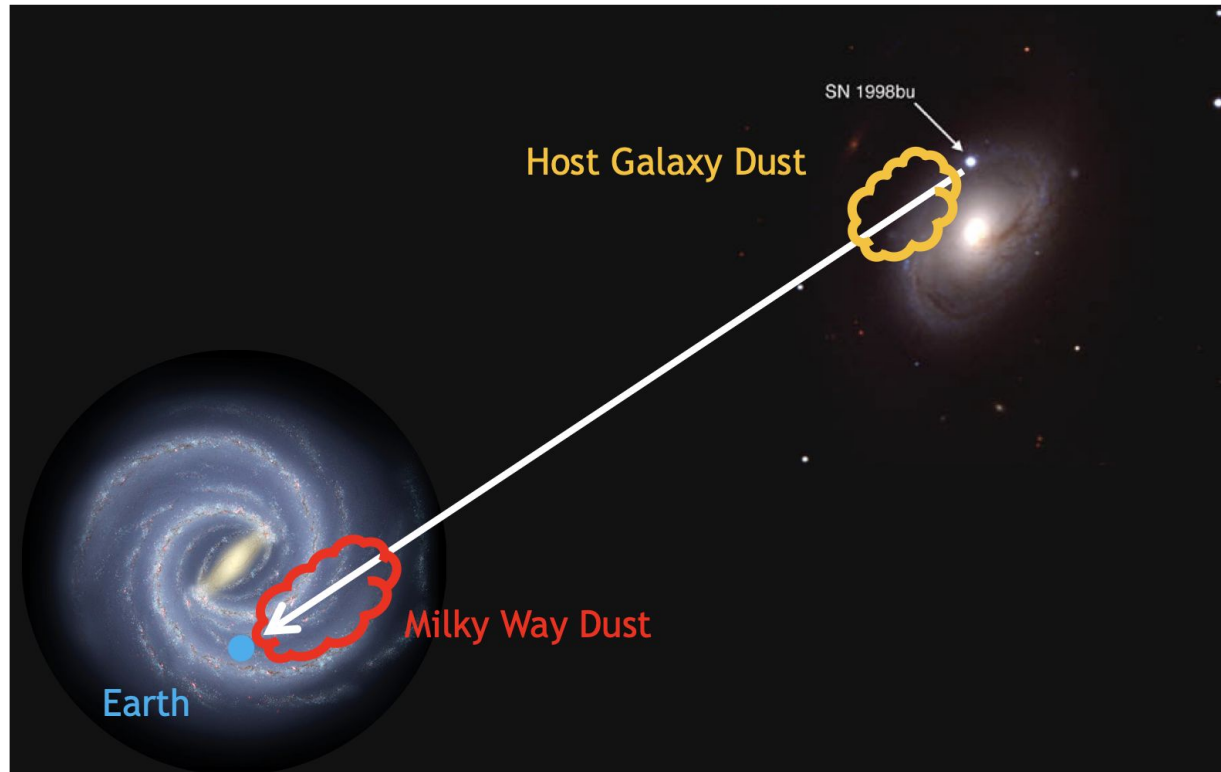
$$\mu_s = m_s - M + \alpha x_s + \beta c_s$$

Diagram illustrating the Tripp formula, showing the relationship between distance, observed brightness, standard brightness, stretch correction coefficient, colour correction coefficient, stretch, and observed B-V colour.

Labels and arrows indicating the variables in the formula:

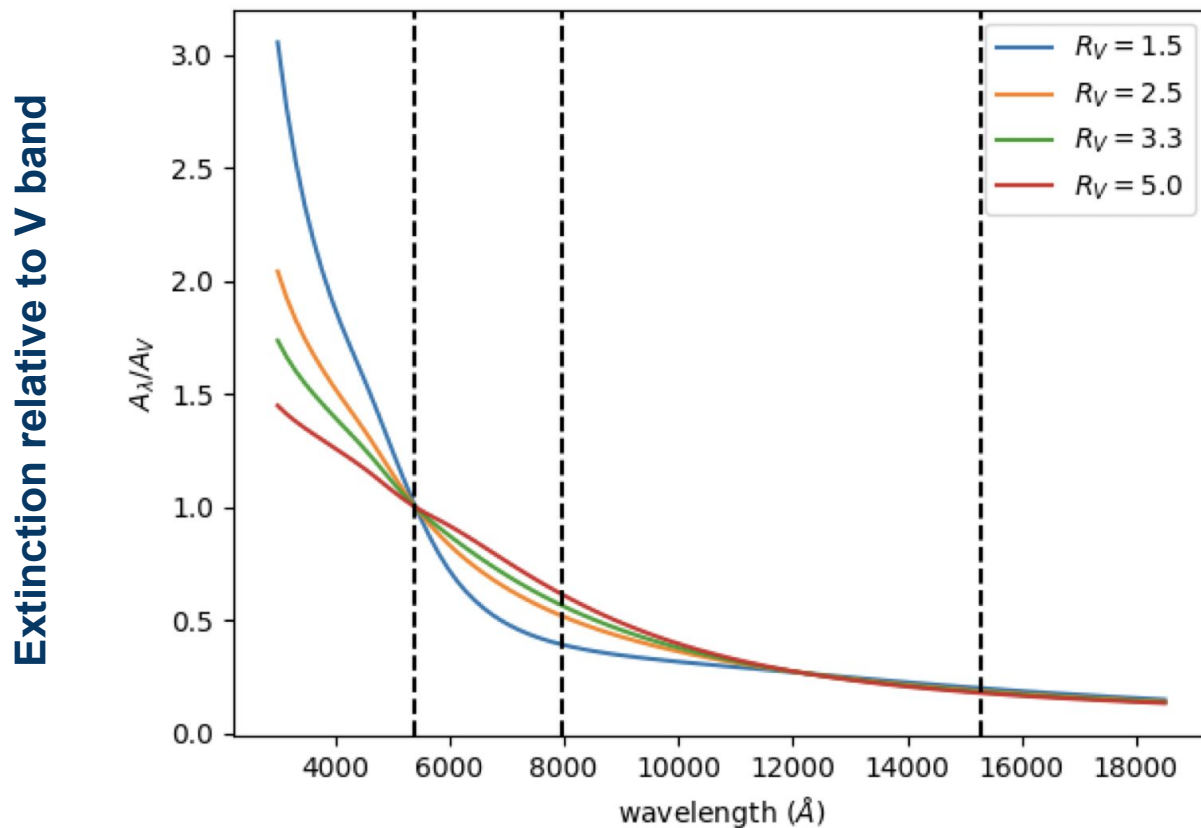
- μ_s : Distance (arrow pointing up)
- m_s : Observed brightness (arrow pointing up)
- M : "Standard" brightness (arrow pointing down)
- α : Stretch correction coefficient (arrow pointing down)
- x_s : Stretch (arrow pointing up)
- β : Colour correction coefficient (arrow pointing down)
- c_s : Observed B-V colour (arrow pointing up)

The Problem? Dust!

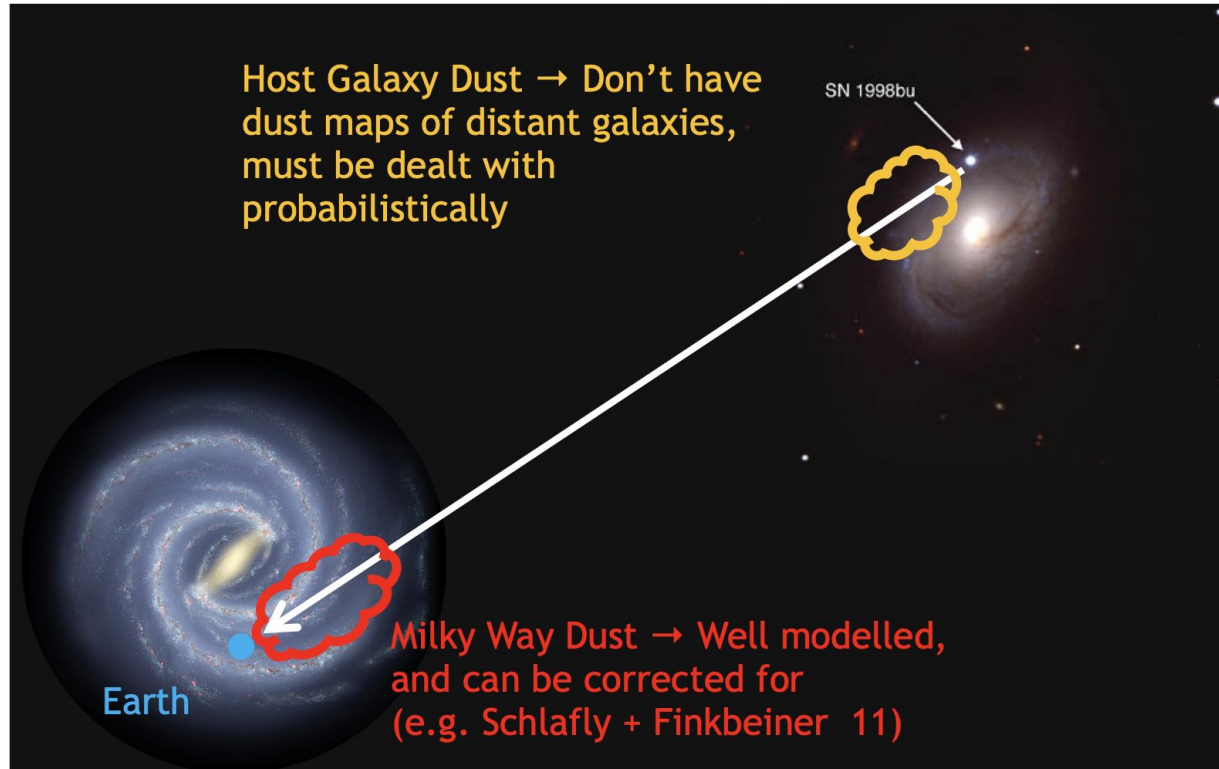


NASA/JPL-Caltech/ESO/R. Hurt
Nicholas B. Suntzeff

Extinction vs. wavelength



Lower $R_V \rightarrow$
steeper dust law



NASA/JPL-Caltech/ESO/R. Hurt
Nicholas B. Suntzeff

$$\mu_s = m_s - M + \alpha x_s + \beta c_s$$

- Tripp formula - one parameter for two separate effects
- Correctly handling dust (and SN Ia colour–luminosity correlations more generally) is key to correctly estimating SN Ia distances
- If intrinsic effect misattributed to dust, could lead to bias

- For typical Bayesian inference, each supernova would be fit separately

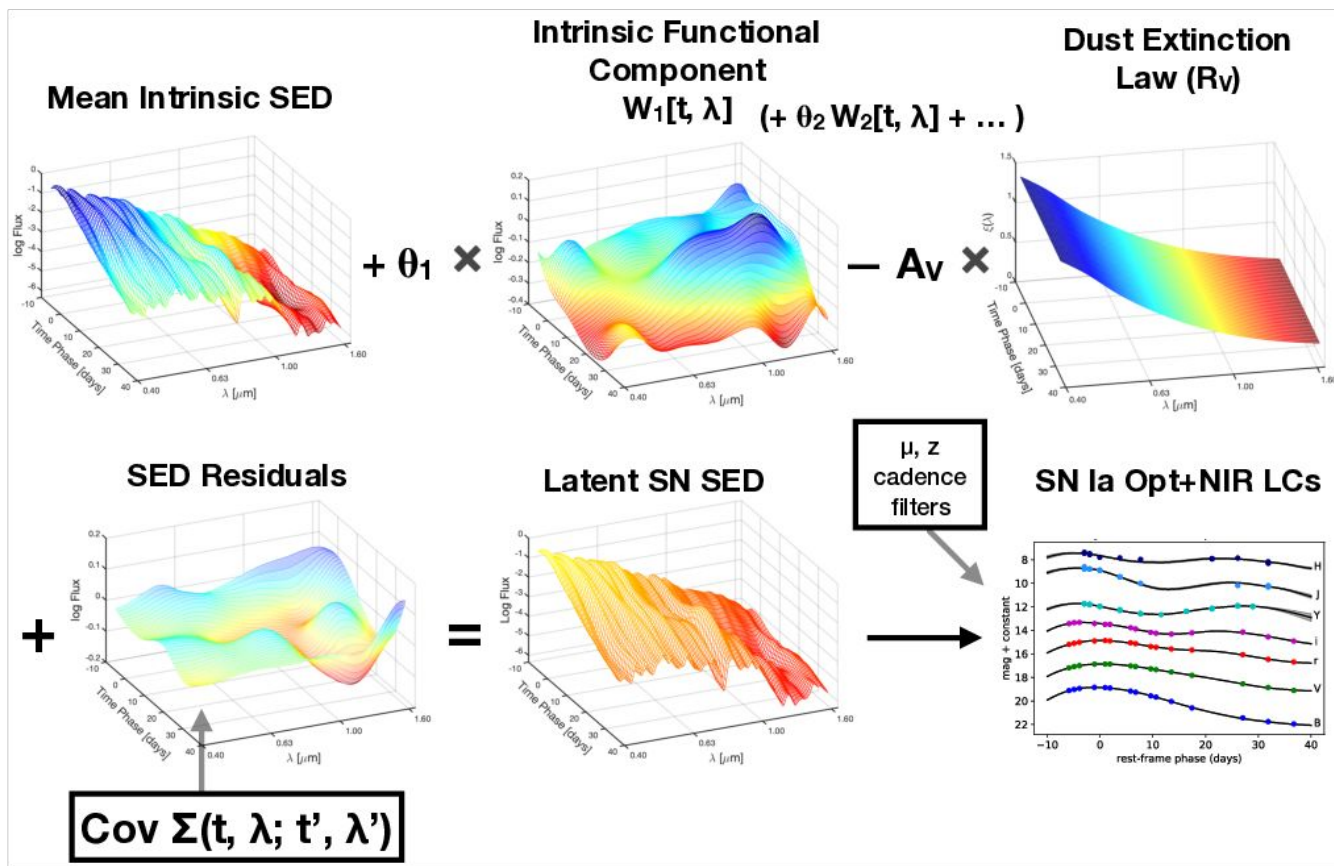
$$R_V \sim U(1, 5)$$

- For a hierarchical model, we jointly sample the posteriors of global and individual supernova properties

$$R_V \sim N(\mu_R, \sigma_R^2)$$

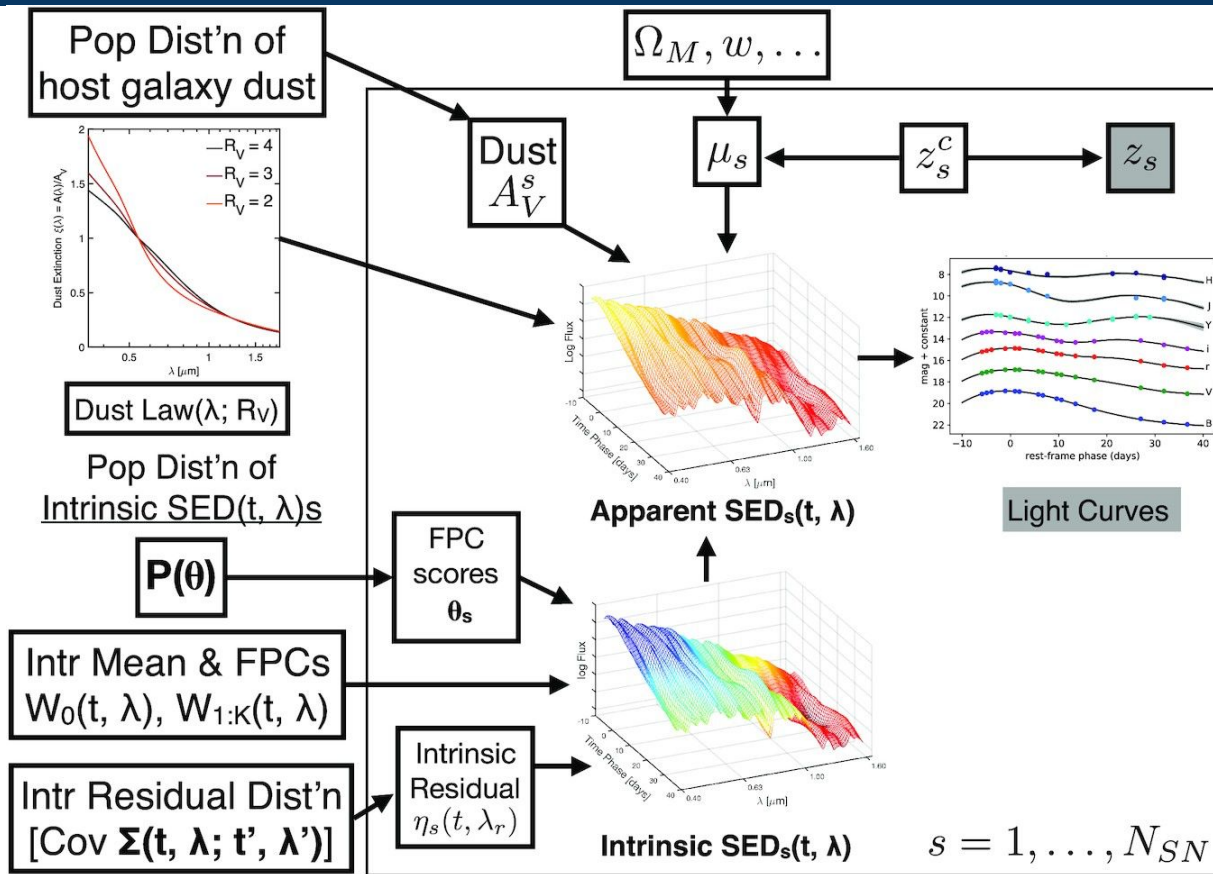
- We can model intrinsic variation and dust properties at the population level as two separate effects

The BayeSN Model

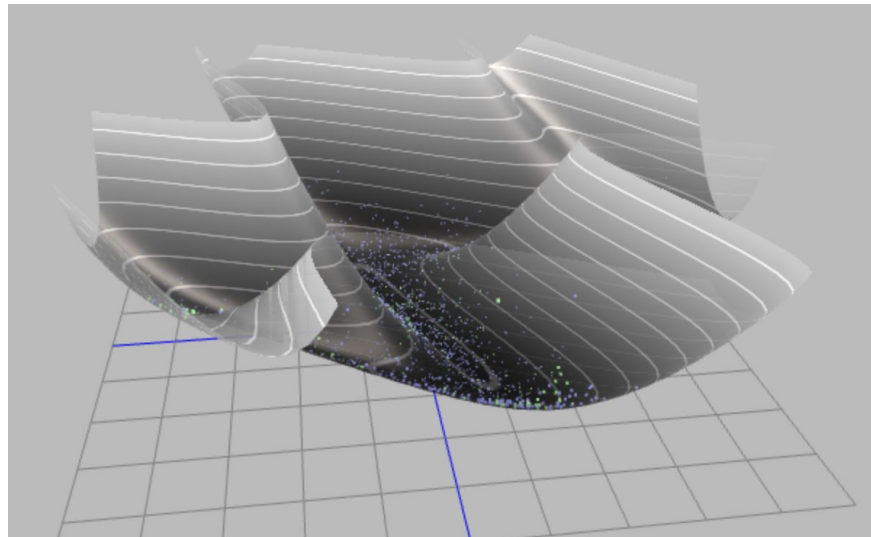


Mandel (2020)

The BayeSN Model



- Two modes of use
 - Training - Conditioning global parameter values on data
 - Fitting - Inference of supernova properties with fixed global parameters
- Uses Hamiltonian Monte Carlo (HMC) to sample posteriors
 - MCMC method using gradient to make efficient steps



Credit: Alex Rogozhnikov

- Hierarchical Bayesian Model
 - Allows for joint inference of global and individual SN properties
 - Better estimation of global R_v distribution using hierarchical approach than just taking mean/standard deviation of individual estimates
 - Can constrain/marginalise over intrinsic SN colour variation
- Model extends into NIR wavelengths, allowing for better constraint of host galaxy dust

- Lots of parameters (e.g. ~ 400 global, ~ 4200 latent parameters for Thorp 2021)
- Complex likelihood evaluation:
 - Compute SED model
 - Evaluate numerical integrals through different filters, for each time of observation and each supernova
- Computationally expensive

Problem:

Scaling BayeSN for next generation data sets without compromising functionality

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Scaling BayeSN for next generation data sets without compromising functionality

Solution:

pyro-ppl/**numpyro**



Probabilistic programming with NumPy powered by
JAX for autograd and JIT compilation to
GPU/TPU/CPU.

 82
Contributors

 529
Used by

 2k
Stars

 179
Forks



Jax

- Python package very similar to numpy, but includes JIT compilation at runtime for any device including GPUs
- Supports autodiff (derivatives with zero computational cost), ideal for HMC
- Very efficient matrix/tensor operations, perfect for numerical integrals

Numpyro

- Probabilistic programming package for Python built on Jax

Vectorized likelihood evaluation + GPUs = Fast Bayesian inference

Mandel+20:

- Trained on 79 *BVriYJH* light curves compiled in Avelino+19
- Previous training ~5 days → now ~20 minutes

Thorp+21:

- Trained on 157 *griz* light curves from Foundation DR1
- Previous training ~1 day → now ~10 minutes

1000 simulated optical SNe:

- Training (conditioning global parameters on data)
 - 45 mins
- Fitting (inference of supernova properties)
 - 15 mins

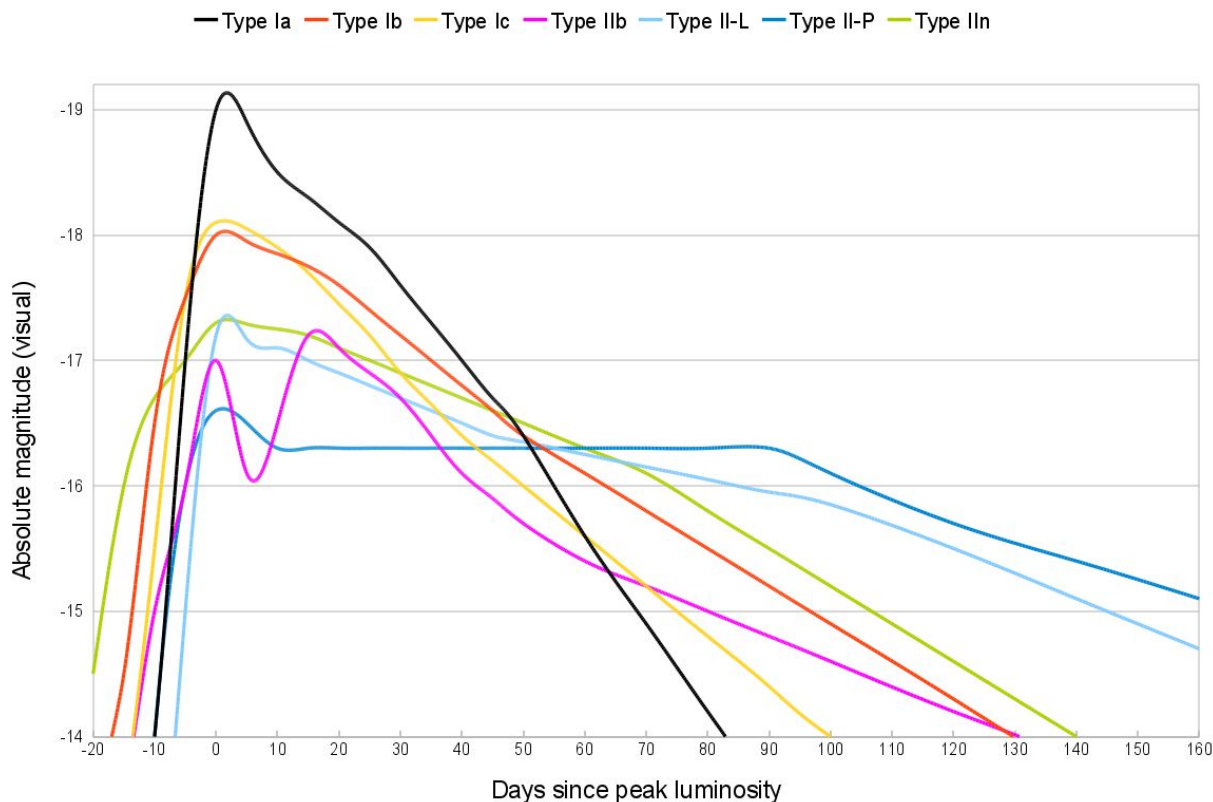
- Previously, training and fitting on data-sets $> 10,000$ SNe would have been computationally unfeasible
- The use of numpyro + GPUs makes this achievable in relatively short timescales
- We are able to scale Bayesian inference approaches to LSST-size data sets
- Further speed increase possible using variational inference (Uzsoy 2022)
- Ambition to have full Bayesian end-to-end analysis from SED level to cosmological parameters

Part 3: Classifying Supernovae

- Supernova classes are based on spectral features
- But...
- LSST will observe far more supernovae than we can obtain spectra for

New Classification Methods are Required

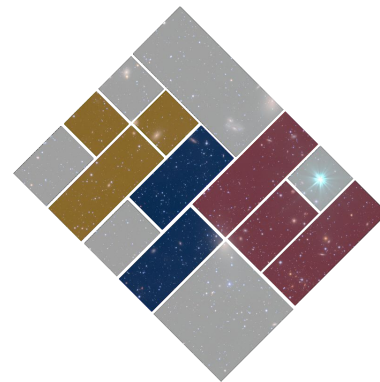
Photometric Classification



- Various machine learning techniques have been applied
 - Convolutional Neural Networks
 - Recurrent Neural Networks
 - Variational Auto-encoders

Young Supernova Experiment (YSE)

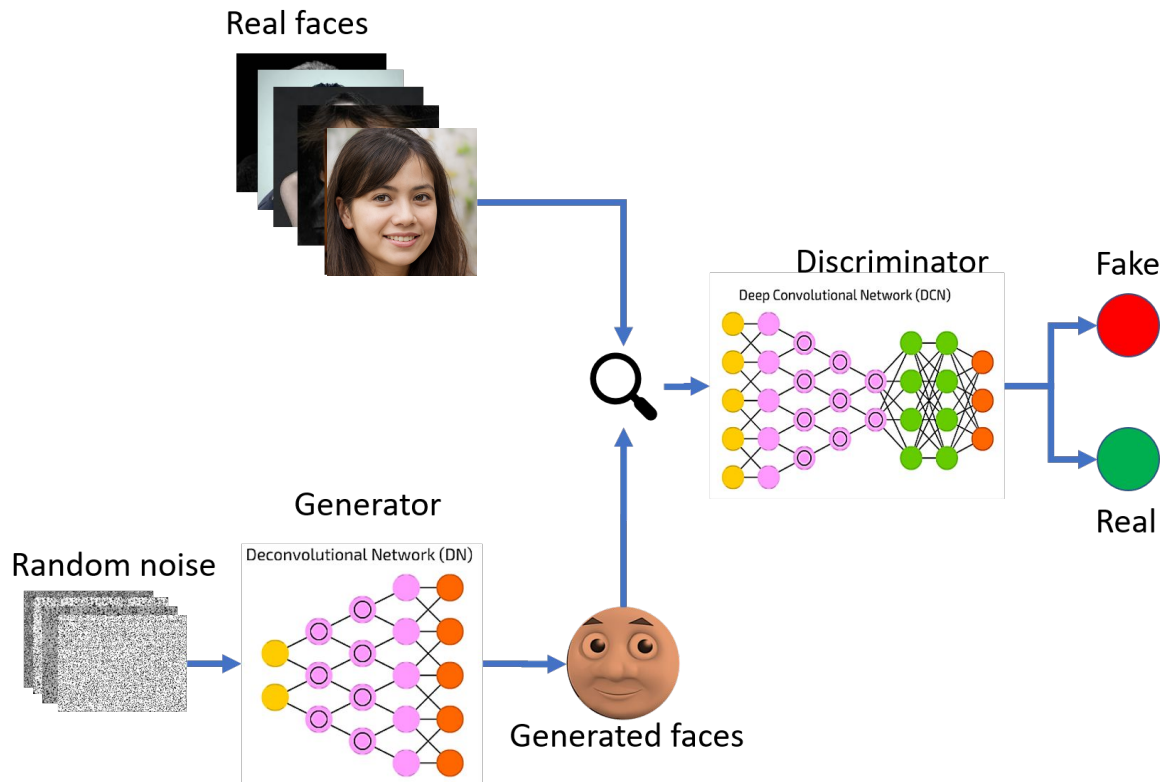
- On Pan-STARRS telescope, currently observing 1500 square degrees of sky every 3 days
- YSE DR1 - Aleo 2023
1975 low redshift supernovae, largest low-redshift, multiband supernova data release ever
- Makes extensive use of photometric classification, ideal preparation for LSST



- Challenges with training on real data
 - Lack of training data
 - Biased training set
- Training on simulated data mitigates for the issues involved
- Limited by quality of simulations
 - Accurate simulations require good physical understanding of population
 - Supernovae are complex and diverse, simulations fail to reproduce full variability of population
 - Consistent drop in performance when applying models trained on simulations to real data

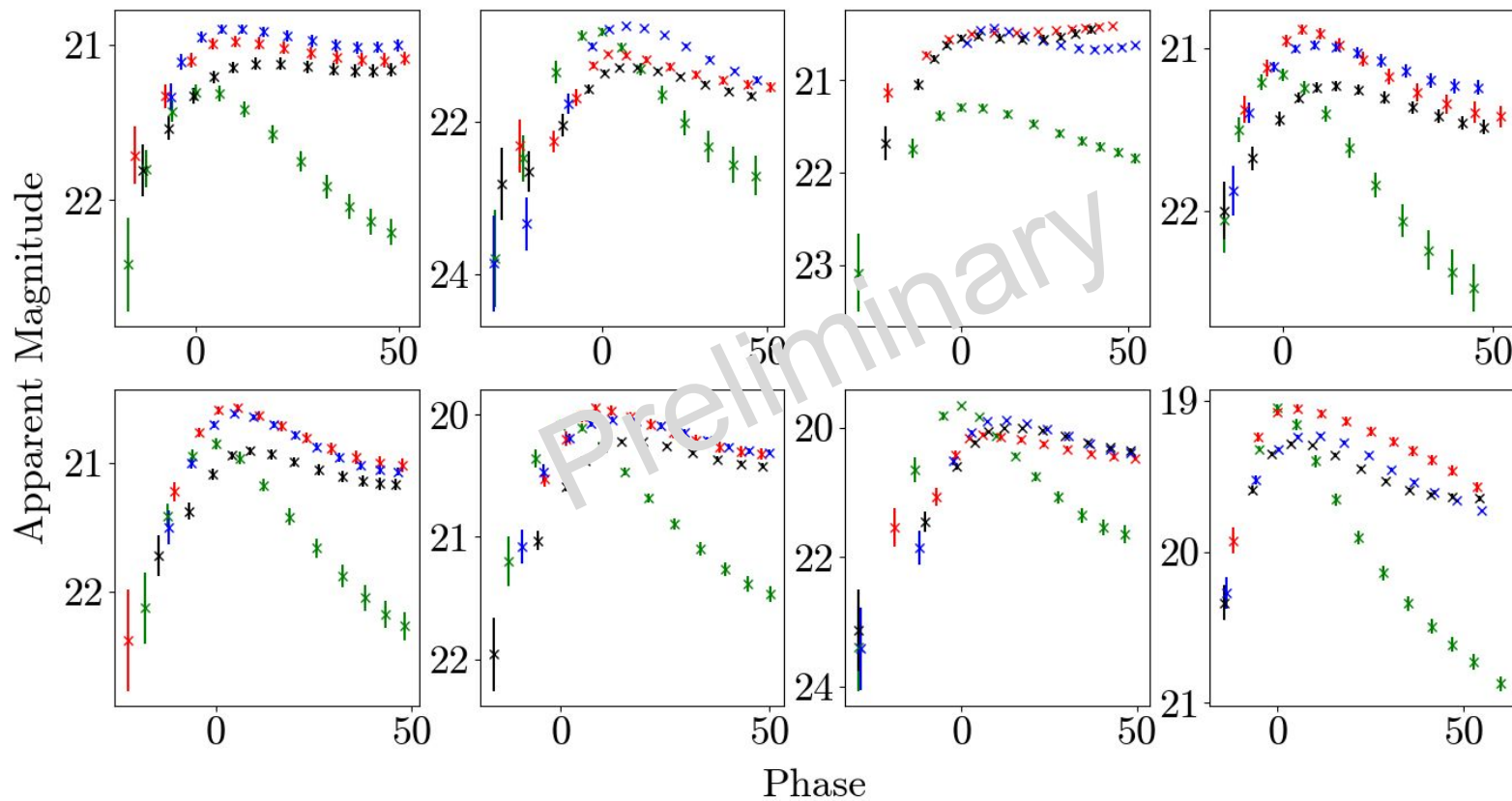
Generative models allow us to sample from the distribution of supernovae, without assuming any physics

Generative Adversarial Network (GAN)

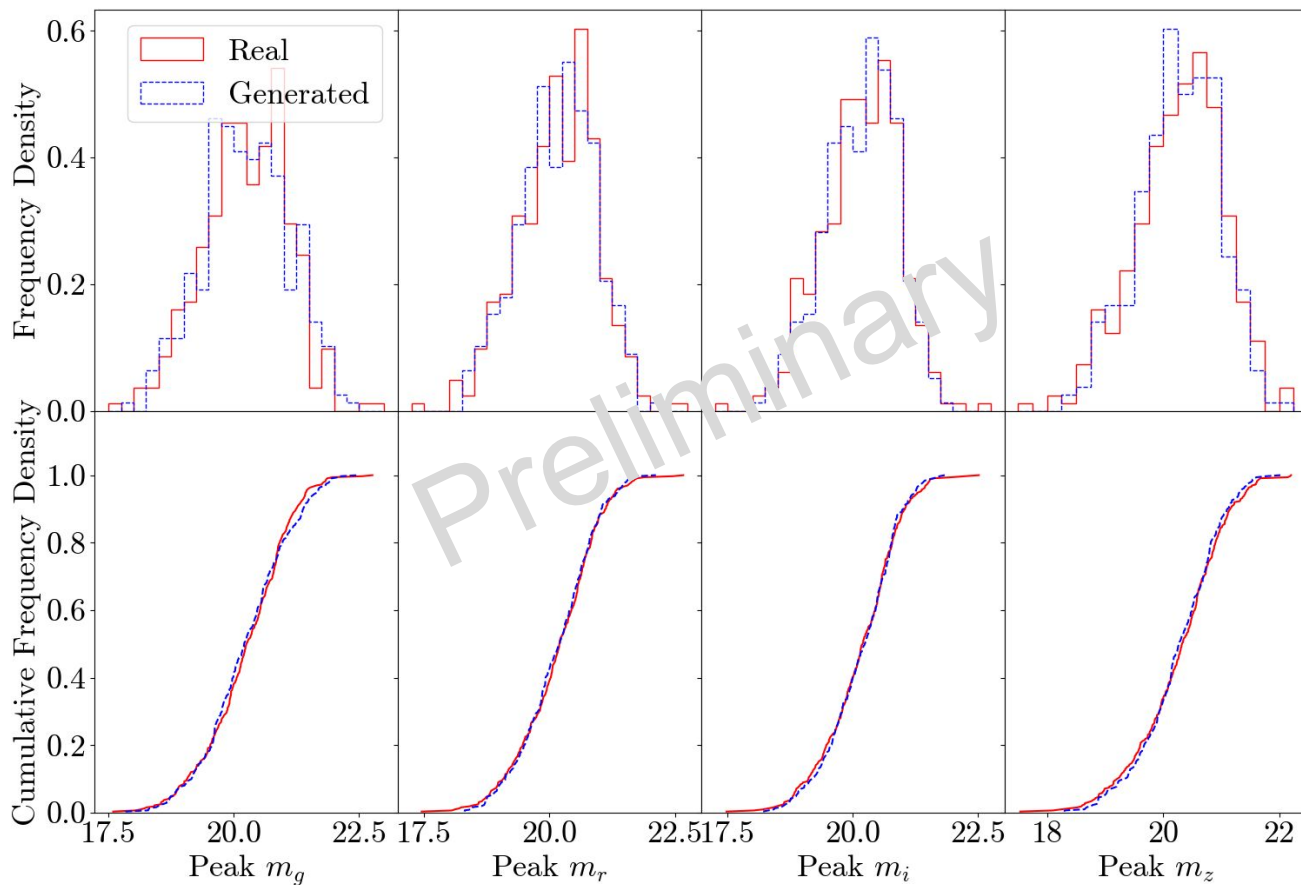


- GANs provide a data-driven approach for generating supernova light curves, able to learn from data without any assumptions about the physics involved
- A recurrent neural network-based GAN is able to generate variable length time series, vital for realistic supernova generation
- Augmenting data-sets using GANs has been shown to improve classification performance within astronomy and beyond (e.g. Motamed+21, Garcia-Jara+22)

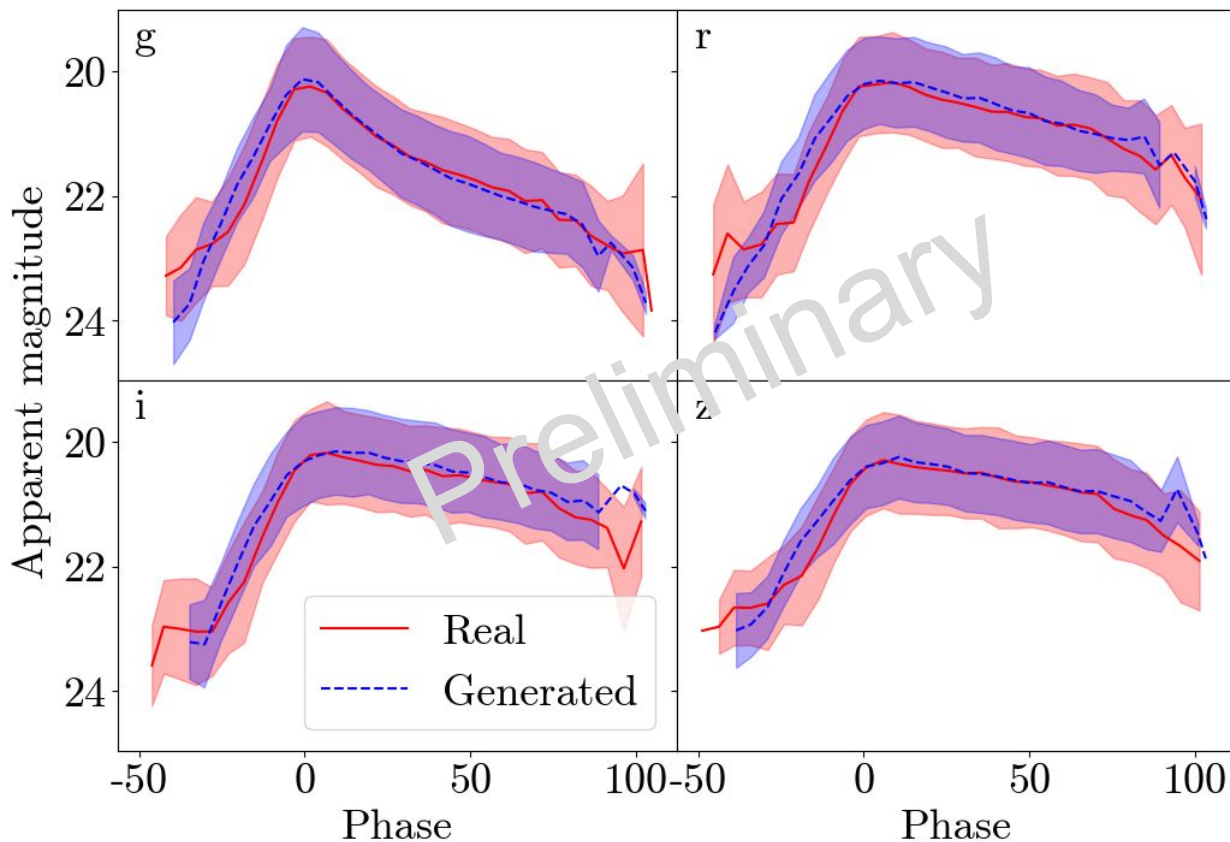
Generated Light Curve Examples



Peak Observed Brightness

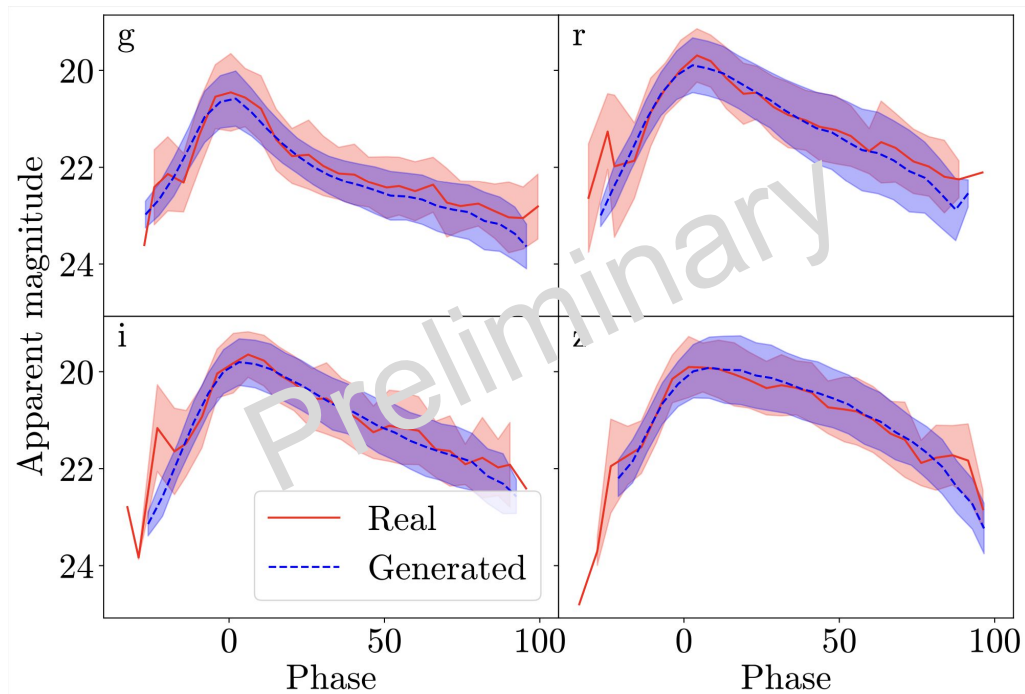


Light Curve Populations



Augmenting Rare Supernova Classes

- Previous plots were for sample of ~350 simulated supernovae
- Training on a much smaller data set of ~30 supernovae still produces comparable results
- For rare SN classes with few examples, this approach has the potential to augment very sparse training sets



- GANs have the ability to generate realistic supernova light curves
- Improvements and future work:
 - Train a classifier with generated data and assess performance
 - Retrain on real data
 - Explore conditional models, allowing one model to generate all types of supernovae

Conclusions

- 1) Supernovae are of great interest within both astrophysics and cosmology, and new surveys pose both opportunities and difficulties for their analysis
- 2) The use of GPU acceleration allows for order-of-magnitude performance increases for Bayesian inference, enabling us to scale BayeSN for next generation surveys
- 3) Generative models such as GANs are able to create physically realistic synthetic supernovae and have the potential to increase the performance of classification models